Adult Income: Model Comparison

Predictive performance, computation cost and categorical handling | 16 October 2026

Principal finding
LightGBM had the strongest observed test scores overall. XGBoost performed very closely and was the model selected by validation. CatBoost was competitive and had the lowest measured batch prediction time, but it was not the predictive winner in this experiment.

Experimental design

The study used a 32,561-row Adult Income dataset and removed 24 exact duplicate records before splitting. The fixed splits contained 22,775 training, 4,881 validation and 4,881 test records. The target was whether income exceeded $50K; 24.09% of test records were positive.

All models used the same 12 original predictors: five numeric and seven categorical, after removing redundant education text and the census weight fnlwgt. Random Forest used one-hot encoding; XGBoost, LightGBM and CatBoost used native categorical inputs. Numeric imputation and categorical Missing handling were learned from training. Each algorithm received a nominal ten-minute Optuna search. Only tuned models were compared; no CV or baseline experiment was included.

Held-out test performance

Model	ROC-AUC	AP	F1	Accuracy
LightGBM	0.9231	0.8251	0.7211	86.64%
XGBoost	0.9228	0.8242	0.7137	86.13%
CatBoost	0.9211	0.8219	0.7127	86.46%
Random Forest	0.9127	0.8024	0.7034	84.33%

Accuracy and F1 here use thresholds selected to maximize validation F1, not a common 0.5 cutoff. Random Forest had the highest recall and balanced accuracy at its chosen threshold, so no model led every possible metric.

LightGBM recorded the highest test AUC, AP and F1, while XGBoost was selected using validation AUC. The validation-based selection was fixed before test evaluation; the test ranking is a descriptive comparison.

What the metrics mean

The models estimate the probability of income exceeding $50K. The evaluation measures ranking quality, probability accuracy and classification decisions.

Metric	Interpretation	Observed result
ROC-AUC	How often a positive record ranks above a negative one; not classification accuracy.	LightGBM 0.9231; XGBoost 0.9228.
Average precision (AP)	Summarizes precision-recall performance across thresholds. It is not trapezoidal PR-AUC.	LightGBM 0.8251; XGBoost 0.8242; CatBoost 0.8219.
F1	Balances precision and recall at the frozen validation-selected threshold.	LightGBM 0.7211 was highest.
Precision	Of records predicted >50K, how many actually were >50K?	CatBoost was highest at 72.89%.
Recall	Of all actual >50K records, how many were identified?	Random Forest was highest at 77.13%.
Log loss / Brier	Probability errors; lower is better. These assess more than calibration alone.	LightGBM was lowest: 0.2864 / 0.0903.
Accuracy / balanced accuracy	Overall correct fraction / average recall across the two actual classes.	LightGBM led accuracy; RF led balanced accuracy (0.8187).

Why Random Forest found more positives but had lower F1

Random Forest flagged 28.74% of records, versus 23.81% for LightGBM. Its broader positive predictions achieved higher recall but lower precision (64.65% versus 72.55%). This is a decision trade-off, rather than an inconsistent result.

Top-10% ranking capacity

Ranking by predicted probability, LightGBM captured 39.29% of >50K records in the top 10%; CatBoost captured 39.20%, XGBoost 39.12%, and Random Forest 38.44%. These boosting differences were small. This is a generic ranking measure, not a proposed targeting policy.

An all-negative classifier would have about 75.91% accuracy but zero recall for >50K. Reporting accuracy alone would therefore miss important behavior.

Uncertainty in model differences

A paired bootstrap compares models on the same resampled test records. No model is retrained. The intervals describe this test sample for fixed models, not new training seeds or new datasets.

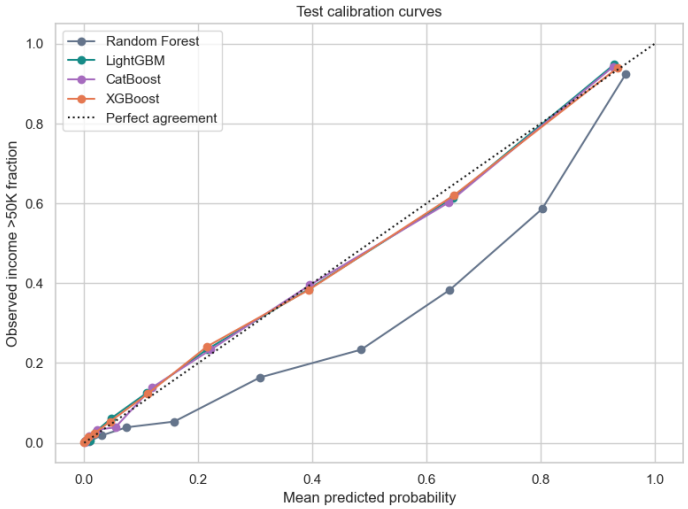
Comparison	Test AUC difference	Paired 95% interval
LightGBM minus Random Forest	+0.0104	[+0.0078, +0.0131]
CatBoost minus Random Forest	+0.0084	[+0.0057, +0.0116]
XGBoost minus Random Forest	+0.0101	[+0.0071, +0.0133]
XGBoost minus LightGBM	-0.0003	[-0.0015, +0.0010]
CatBoost minus LightGBM	-0.0020	[-0.0036, -0.0005]
XGBoost minus CatBoost	+0.0017	[+0.0001, +0.0035]

Boosting versus Random Forest: all three paired AUC intervals were positive, supporting stronger discrimination for these boosting pipelines in this test analysis.

LightGBM versus XGBoost: the interval crossed zero. The experiment does not clearly separate them in AUC; this is not proof of exact equivalence.

CatBoost: LightGBM's AUC advantage was supported by this bootstrap, and XGBoost's AUC advantage had a lower bound close to zero. LightGBM's AP advantage over CatBoost also had a narrowly positive interval. These are exploratory, pointwise intervals, without multiple-comparison adjustment.

Probability calibration



Boosting curves followed the diagonal fairly closely at the ten plotted bin averages. Random Forest's curve lay below it through much of the range, indicating overprediction there. Its selected balanced class weighting may contribute, but this experiment does not isolate that cause. Bin averages do not prove perfect calibration for every subgroup.

Fit cost versus prediction cost

Computation times describe the fitted configurations on the study machine. Inference used the same validation batch size, warm-up, seven repeats and an explicit four-thread setting for each API.

Model	Final fit seconds	Prediction sec / 1,000	Model size (MiB)
Random Forest	2.7493	0.0816	17.0052
LightGBM	1.5107	0.0560	2.3594
CatBoost	36.1999	0.0088	0.8280
XGBoost	1.1946	0.0095	0.7875

XGBoost fitted fastest in this Adult experiment. CatBoost had the lowest measured median batch inference time, although its roughly 0.0007-second gap from XGBoost per 1,000 records is small and does not establish a robust general speed advantage. XGBoost had the smallest saved model file. Random Forest was the largest.

These are model-only prediction times: loading and preprocessing are excluded. A 1,000-record batch time is not a single-record latency. Fit timings were single recorded refits; file size is not runtime RAM usage.

Search coverage under the time budget

Model	Completed trials	Recorded search-session seconds
Random Forest	153	605.18
LightGBM	331	600.25
CatBoost	18	624.67
XGBoost	372	601.71

CatBoost explored only 18 configurations compared with hundreds for LightGBM and XGBoost. That is a meaningful result under fixed-time tuning, but it does not establish CatBoost's maximum achievable quality. Trial durations exclude some optimizer/database overhead; an ongoing trial can overrun the nominal 600-second budget.

Operational trade-offs
LightGBM is the observed score leader; XGBoost combines very similar scores with a fast fit, small model and fast inference; CatBoost offers competitive quality and fast observed inference at a higher final-fit cost. The pre-test validation selection remained XGBoost.

CatBoost and categorical features

CatBoost is designed to handle categorical variables well. That is a useful capability, not a guarantee of the highest score on every categorical-heavy dataset.

What native handling means

CatBoost received categorical strings through cat_features, without an external one-hot encoding stage. CatBoost can build numeric statistics and combinations from categories. Its research describes techniques for reducing target leakage and prediction shift in categorical processing. For small-category features it can still use one-hot encoding internally [1, 2].

Factors relevant to the observed ranking

Its competitors also handled categories natively. All three boosting pipelines used their native categorical support.
Column count is not predictive strength. Seven categorical predictors do not guarantee they dominate the five numeric predictors.
Budget and settings matter. CatBoost completed fewer trials, and this study did not exhaust its configuration choices.
Every dataset differs. Cardinality, rare categories, interactions and sample size affect which approach performs well. These are plausible influences; the experiment did not isolate their individual effects.

Predictive signals in the fitted CatBoost model

CatBoost's leading input was relationship, followed by capital.gain, age and education.num. Occupation was also prominent. Both categorical and numeric information mattered. These are built-in within-model importances, not causal effects or directly comparable scores across libraries.

Conclusion
On this Adult Income split, LightGBM achieved the highest observed test discrimination, average precision and F1, while XGBoost was not clearly separated from LightGBM in the paired AUC comparison and had been selected by validation. CatBoost handled categorical inputs successfully and predicted quickly, but it did not outperform the competing native boosting pipelines in predictive quality. The result supports testing CatBoost as a strong candidate, not assuming it will win whenever categories are present.

Limits and source notes

Single split and training seed; only tuned pipelines; no within-algorithm one-hot/native ablation; unweighted row evaluation; education text and fnlwgt excluded; exploratory 300-replicate paired intervals. This is a comparison of complete pipelines, not proof that native encoding caused the boosting advantage. Test evaluation reports frozen model choices rather than a further tuning stage.

Results source: Adult Income experimental evaluation records, 16 October 2026. Values are rounded from saved metric tables; the calibration figure is from the same evaluation.

[1] CatBoost official documentation: categorical features
[2] Prokhorenkova et al., CatBoost: unbiased boosting with categorical features, NeurIPS 2018